

**California State Procurement Opportunity Landscape:
A Reproducible Analysis of Cal eProcure Bid Activity**

Luis Muniz

Udacity Institute of AI & Technology

Master's Degree in AI Capstone

Abhishek Mann and Deepak Sharma, Capstone Mentors

July 31, 2026

California State Procurement Opportunity Landscape

Overview

This project created a complete, reproducible Python workflow for analyzing California Cal eProcure proposal-request data. The workflow combined 5,542 completed solicitations with 290 currently open solicitations, producing 5,832 source records for cleaning, exploratory analysis, visualization, and interpretation. Cal eProcure is California's portal for advertised state bid opportunities in the California State Contracts Register (FI\$Cal, n.d.). The analysis was designed both to demonstrate AI programming foundations and to create a practical procurement-monitoring foundation for ICM Solutions.

Dataset Description

The source data consisted of two Cal eProcure exports downloaded on July 24, 2026, from the [Cal eProcure Event Search page](#) (California Department of General Services, n.d.). One export covered completed solicitations with deadlines from July 1, 2025 through June 30, 2026, and the other contained currently open solicitations as of July 24, 2026. The completed file contained 5,542 rows and the open file contained 290 rows; each source file initially contained 10 columns. The core variables included department, department name, event ID, event name, format, type, end date, status, buyer name, and buyer email. The analysis focused on deadline timing, department-level solicitation volume, data-quality conditions, and a conservative keyword screen for current opportunities that might align with ICM Solutions. The files used an .xls extension but were HTML tables, so the ingestion method was selected based on the files' actual content rather than their extension.

Workflow Description

The workflow began by reading each source export with `pandas.read_html()`, validating that one table was present, and adding a record-group field. Cleaning then standardized column names, trimmed whitespace, normalized missing text values, repaired a recurring export character artifact, and preserved identifiers as strings. Deadline strings were converted into datetime values and used to derive month, weekday, calendar date, and days-until-deadline features. A composite record key was created because five Event IDs were reused across different departments in the completed dataset. Exploratory analysis reviewed dimensions, missing values, duplicated identifiers, department counts, and date coverage. The notebook then produced three visualizations and a manually reviewable table of current opportunities that matched conservative ICM service phrases.

The notebook integrates data inputs, code, outputs, visualizations, and narrative explanations in one computational record. This design follows Danchev's (2022) emphasis on transparent and reproducible data-science workflows and Rule et al.'s (2019) recommendation to use notebooks as a documented computational narrative rather than a collection of unexplained code cells.

Key Decisions and Assumptions

The original solicitations were retained when buyer name or buyer email was missing because those fields were not necessary for the main analysis. Dropping those rows would have reduced department-level counts without resolving an analytical requirement. The original deadline text was retained alongside parsed datetime values to support traceability. PDT and PST suffixes were removed before parsing because the analysis compared California local calendar

dates, not timestamps across time zones. Event ID was not treated as globally unique; instead, department, Event ID, and deadline were combined into a record key.

The ICM opportunity screen used only direct phrases connected to cloud services, application support, systems integration, strategic planning, IT training, data integration, and cybersecurity. The rule set was intentionally conservative to avoid matching physical systems, building maintenance, security-guard services, or unrelated consulting. A match was interpreted only as a reason for human review. It was not treated as evidence of eligibility, scope fit, contract value, procurement-vehicle compatibility, or probability of award.

Results and Interpretation

The cleaning and validation process found no deadline-parsing failures and no exact duplicate rows. Forty-four completed records and three open records lacked both buyer name and buyer email. Five Event IDs were reused across different departments, confirming that Event ID alone was not a safe unique key. These findings demonstrate why validation was necessary even though the exports appeared well structured.

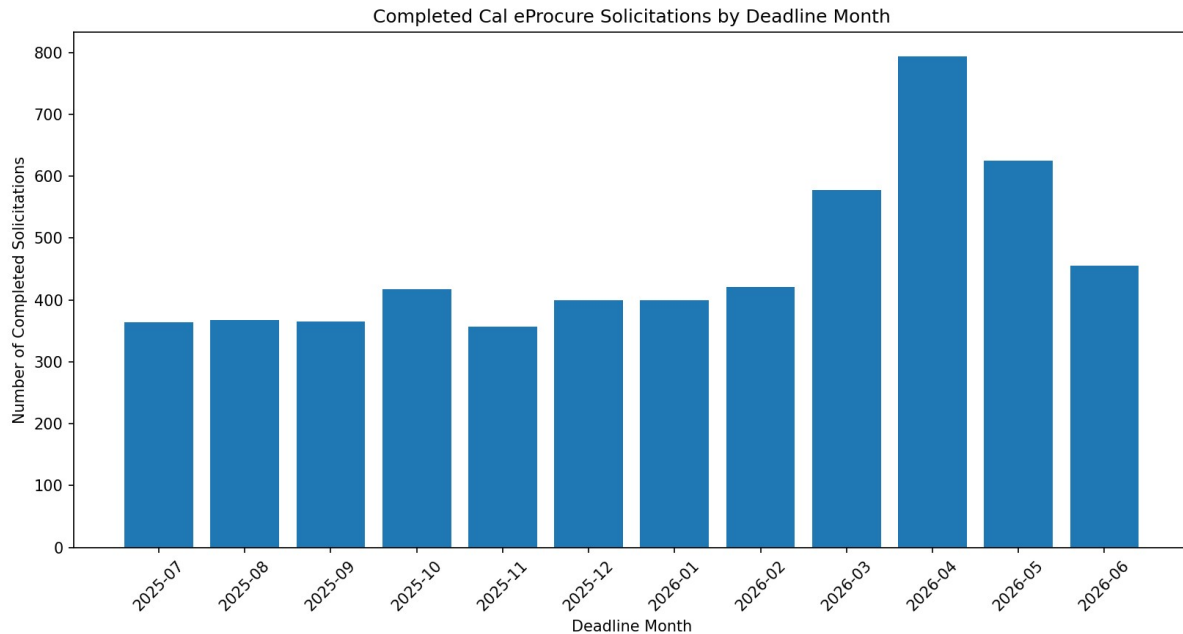


Figure 1. Completed Cal eProcure solicitations by deadline month.

Figure 1 shows that completed solicitation deadlines were highest in April 2026, with 793 records, followed by May 2026 with 625. November 2025 had the lowest monthly total, with 357. The rise in March through May may be consistent with a procurement-cycle effect, but the available fields do not establish why the pattern occurred because publication dates and fiscal encumbrance dates were not provided.

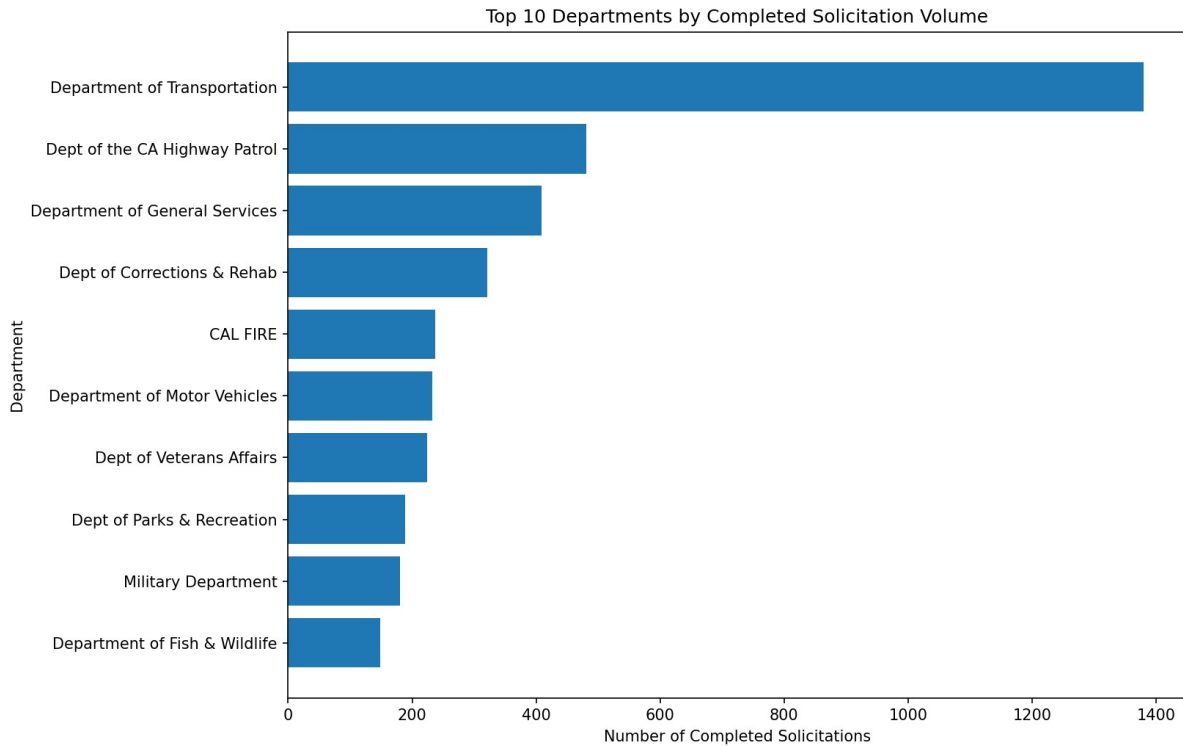


Figure 2. Top 10 departments by completed solicitation volume.

Figure 2 shows that the Department of Transportation accounted for 1,380 of 5,542 completed solicitations, or approximately 24.9 percent. The Department of the California Highway Patrol and Department of General Services followed with 481 and 409 records. This concentration means that statewide volume is heavily influenced by transportation, construction, facilities, and related procurements and should not be interpreted as a direct measure of ICM's technology-consulting market.

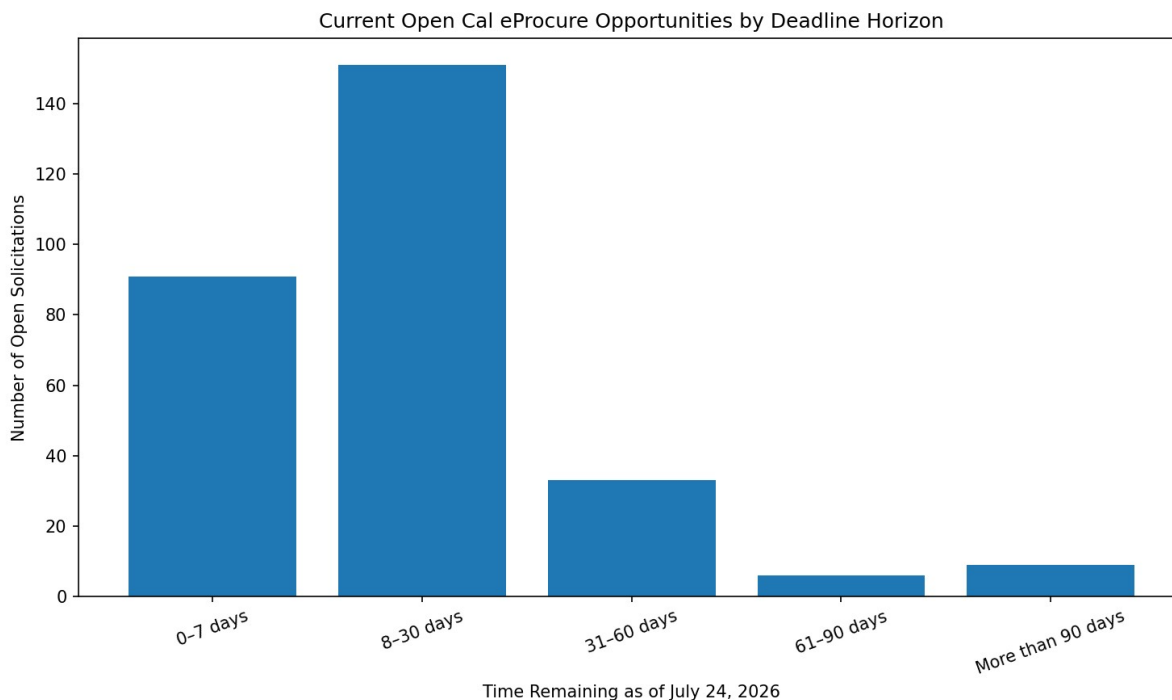


Figure 3. Current open opportunities by deadline horizon as of July 24, 2026.

Figure 3 shows that 91 open solicitations were due within seven days and 151 were due within 8 to 30 days. Together, 242 of 290 open opportunities, or approximately 83.4 percent, were scheduled to close within 30 days of the snapshot. This result supports frequent monitoring, but it does not measure the full response window because publication dates were absent.

The conservative ICM phrase screen flagged 14 current event titles for manual review. Examples included EDD Salesforce maintenance and operations, online or virtual IT training, enterprise strategic planning, managed cybersecurity services, application support, cloud services, data integration, and systems-integration opportunities. These titles illustrate potential business relevance, but every opportunity requires review of the authoritative solicitation package before resources are committed.

Responsible Practice (Bias and Data Quality)

Cleaning decisions could have produced misleading results if missing buyer contacts had been treated as invalid records, if reused Event IDs had been collapsed, or if text artifacts had been silently changed without retaining the original fields. The workflow reduced these risks by preserving solicitations with missing contacts, using a composite record key, retaining original deadline text, and documenting the specific text normalization performed. The notebook also makes the difference between observed patterns and unsupported explanations explicit.

The ICM screen creates a separate source of bias because it reflects the language selected by the analyst. It can miss relevant solicitations that use unfamiliar terminology and can flag titles whose complete scopes are not aligned. Danchev (2022) emphasizes that reproducible data science includes critical consideration of selection bias, data ethics, and the social impact of analytical choices. For this project, the appropriate safeguard is to keep the screening rules visible, describe the results as leads rather than recommendations, and require human review.

Reproducibility

The repository retains the two original exports in a raw-data folder and uses relative file paths so the notebook can be run after cloning. The notebook creates the processed combined CSV and all figure files rather than relying on manually edited outputs. The requirements.txt file records the Python dependencies and was generated from the activated project environment using pip freeze. Git commits and a development branch document the progression from ingestion through cleaning, exploratory analysis, visualization, interpretation, and final validation.

Danchev (2022) explains that dependency files help reproduce the software environment used by computational notebooks. Rule et al. (2019) further recommend organizing notebooks

into clear steps, documenting the process and reasoning, and ensuring that the notebook can be rerun in a predictable order. During final validation, every notebook cell was executed from top to bottom, no traceback occurred, and the processed dataset and all three figures were recreated successfully.

References

FI\$Cal. (n.d.). Cal eProcure resources. <https://fiscal.ca.gov/user-support/cal-eprocure-resources/>

California Department of General Services. (n.d.). Cal eProcure event search [Data set].

Retrieved July 24, 2026, from <https://caleprocure.ca.gov/pages/Events-BS3/event-search.aspx>

Danchev, V. (2022). Reproducible data science with Python: An open learning resource. *Journal of Open Source Education*, 5(56), 156. <https://doi.org/10.21105/jose.00156>

Rule, A., Birmingham, A., Zuniga, C., Altintas, I., Huang, S.-C., Knight, R., Moshiri, N., Nguyen, M. H., Rosenthal, S. B., Pérez, F., & Rose, P. W. (2019). Ten simple rules for writing and sharing computational analyses in Jupyter Notebooks. *PLOS Computational Biology*, 15(7), e1007007. <https://doi.org/10.1371/journal.pcbi.1007007>